Alexandre Autran

Normalien in Mathematics and Machine Learning

Cesson-Sévigné, France alexandre.autran@ens-rennes.fr +33 7 68 91 72 04

Personal Webpage LinkedIn (+2k followers) GitHub Google Scholar

Profile

I am a machine learning and mathematics student at the École Normale Supérieure (ENS) in Paris-Saclay and Rennes. My research focuses on probabilistic and geometric methods for machine learning, stochastic processes and stochastic control, high-dimensional statistics, density analysis, optimal transport, graph-based learning, and statistical learning theory.

Education

2024–2027 **ENS Diploma, Department of Mathematics**
Admitted to the ENS through a competitive entrance examination.

2026–2027 **Research year in Machine Learning**, *École Normale Supérieure Paris-Saclay*, Paris, France
Final year of the ENS Diploma (research track at ENS de Rennes).

2025–2027 **MSc in Mathematics**, *École Normale Supérieure de Rennes*, France
MSc in Mathematics (M2). Highest honors (17/20), first semester.
MSc in Mathematics (M1). Highest honors (17.6/20). Rank: 5th/60+.

2024–2025 **BSc in Mathematics, Computer Science minor**, *École Normale Supérieure de Rennes*, France
High honors (15.4/20).

Publications

[4] **A. Autran**, “Improved Wasserstein Estimates for Jump-Diffusion Perturbations under Semigroup Smoothing”, preprint, 2026.

[3] **A. Autran**, “Second-Order Asymptotics of Emergent α-Stable Common Noise”, preprint, 2026.

[2] **A. Autran**, “Optimization with an Exact Discrete Adjoint for a Diffusive Lotka–Volterra Harvesting Problem”, preprint, 2026.

[1] **A. Autran**, D. Munteanu, and J.-L. Autran, “Multi-Dimensional Charge Diffusion–Collection Model in Semiconductor Devices Subjected to Single Ionizing Particles”, *Applied Sciences*, vol. 16, no. 11, Art. no. 5551, 2026. [Paper]

Research Experience

Aug 2026– **Machine Learning Researcher**
High-Dimensional Density Analysis Using Local Unimodality.
Supervised by Argyris Kalogeratos (Senior Researcher) at Centre Borelli, ENS Paris-Saclay, France.

May–Aug 2026 **MSc Research Internship**
Optical Modeling of Thermochromic VO_2-Based Oxide Nanocomposites [Report] [Presentation]
Supervised by Prof. Nguyen Ngoc Duy and Amaury Baret (PhD Student) at the University of Liège, Belgium.

Apr–Jun 2026 **MSc Research Internship**
Numerical Simulation and Optimal Control of Spatial Lotka–Volterra Systems. [Report]
Supervised by Prof. Martin J. Gander and Bastien Chaudet-Dumas (Postdoctoral Researcher) at the University of Geneva, Switzerland.

Apr–May 2026 **Machine Learning Projects**
Cardiovascular Disease Prediction Using Statistical Learning and Neural Network Models; Air Pollution Prediction from Environmental Data. [Cardiovascular report] [Air-pollution report]
Studies carried out as part of the MSc course *Statistical Learning* (Prof. Myriam Maumy).

Jan 2026 **Statistical Reading Group**
Supervised Statistical Learning – Prediction by Empirical Risk Minimization. [Presentation]
Supervised by Prof. Magalie Fromont at ENS de Rennes, France.

Nov–Dec 2025 **MSc Research Seminar**
Point Process and Stein's Method. [Report] [Presentation]
Supervised by Prof. Jean-Christophe Breton at the University of Rennes, France.

Oct–Nov 2025 **Algebra Reading Group**
Elimination Theory and Extension. [Presentation]
Supervised by Jade Nardi (Research Scientist) at *ENS de Rennes*, France.

Jun–Aug 2025 **BSc Research Internship**
Lattice-Based Post-Quantum Cryptography. [Report] [Presentation]
Supervised by Prof. Pierre-Alain Fouque and Aurore Guillevic (Research Scientist) at Inria, France.

May–Jun 2025 **BSc Research Internship**
Algebraic Counting of the Real Roots of Polynomials. [Report] [Presentation]
Supervised by Prof. Marie-Françoise Roy at the University of Rennes, France.

Feb–Mar 2025 **Directed Research Reading**
Geometry of Numbers. [Report] [Presentation]
Supervised by Juliette Bavard (Research Scientist) at ENS de Rennes, France.

Feb–Mar 2024 **BSc Research Internship**
Topological Study of Matrix Groups. [Report]
Supervised by Prof. Françoise Dal'Bo and Jérémy Bettinger (PhD Student) at the University of Rennes, France.

May–Jun 2023 **BSc Research Internship**
Introduction to Hyperbolic Geometry and Fuchsian Groups. [Report] [Presentation]
Supervised by Frank Loray (CNRS Research Director) at the University of Rennes, France.

Other Activities

2024–2025 **ENS Oral Examinations**
Supervision of 2nd year BSc students in preparation for the oral entrance examinations in mathematics of the Écoles Normales Supérieures: Ulm, Paris-Saclay, Rennes and Lyon. [Resources]

2022–2025 **Teaching in Mathematics**
Tutoring for final-year high school students of the Lycée Chateaubriand in Rennes, and for first and second years BSc students of the University of Rennes. [Resources]

2023–2024 **BSc Student Representative**
Representative on the council of the Mathematics Department of the University of Rennes.

Before 2022 **High School Competitions**
Concours Général in Mathematics, Physics, Philosophy, and National High School Mathematics Olympiad at Lycée Thiers in Marseille.

Computer Skills

Programming: Python, C/C++, Java, MATLAB, R.
Python / Machine Learning: NumPy, pandas, scikit-learn, PyTorch.

Languages

French: native language. **English:** professional working proficiency.

Sports and Interests

Sports: running, cycling, strength training.
Interests: reading and writing, in fantasy and science fiction literature; chess.

Academic Reference

François Bolley [Web page]
Full Professor and Director of the Mathematics Department at ENS de Rennes.
Email: francois.bolley@ens-rennes.fr.